

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q15-0540

Lincoln Park 100 kV  
November 24, 2017

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0540. The Interconnection Customer proposes a 334.3 MW natural gas combined cycle facility interconnecting at the Lincoln Park 100 kV substation in the Ohio area. The request was received on November 18, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 195 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,172,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                      | Contingency                              | Loading (%) | Rating (MVA) | DFAX (%) |
|-----------------------------------------|------------------------------------------|-------------|--------------|----------|
| LINCOLN PARK / KOSCIUSKO 100 KV L       | LINCOLN PARK / TURNING STONE 100 KV      | 232.5       | 112          | 63.7     |
| TURNING STONE / LINCOLN PARK 100 KV     | LINCOLN PARK / KOSCIUSKO 100 KV L        | 149.0       | 142          | 43.8     |
| TURNING STONE / BELLE VISTA 100 KV Line | LINCOLN PARK / KOSCIUSKO 100 KV          | 142.6       | 130          | 43.8     |
| MOSIER / KOSCIUSKO 100 KV               | TURNING STONE / LINCOLN PARK 100 KV Line | 118.3       | 130          | 34.9     |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                            | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-------------------------------------|---------------------------------------------|---------------|-----------------------|
| LINCOLN PARK / KOSCIUSKO 100 KV L   | Rebuild line with higher-capacity conductor | \$2,487,000   | 100% (\$2,487,000)    |
| TURNING STONE / LINCOLN PARK 100 KV | Rebuild line with higher-capacity conductor | \$2,621,000   | 100% (\$2,621,000)    |

| Facility                           | Required Upgrade                            | Cost Estimate | Allocation to Project |
|------------------------------------|---------------------------------------------|---------------|-----------------------|
| TURNING STONE / BELLE VISTA 100 KV | Rebuild line with higher-capacity conductor | \$4,278,000   | 100% (\$4,278,000)    |
| KOSCIUSKO / MOSIER 100 KV          | Rebuild line with higher-capacity conductor | \$2,786,000   | 100% (\$2,786,000)    |
|                                    | Total allocated to project                  |               | \$12,172,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.